Agriculture and the Environment

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Agriculture and the Environment (A32001)

Short Title:	Agriculture & the Environment		
Faculty	Science and Computing		
Department	Science		
Cluster	Environmental Science		
Author	TWOODCOCK		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module is designed to equip students with a knowledge of the impact of farm practices on the environment, including the impacts of slurries, fertilisers and pesticides. The student will undertake a basic Environmental Impact Assessment of a site and write an Environmental Impact Statement. Students will learn about sustainable agriculture and carry out laboratory analysis of pollutants in water and soil. Students will discuss the potential advantages and disadvantages of incorporating clover into grass swards. New EU and Irish legislation and schemes relating to farm practices and their impact on the environment will be discussed. Students will visit an on-farm constructed wetland system and analyse water samples in the lab to test the effectiveness of this water purification technique.

Programmes

AGRI-0015	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	3 / 5 / M
AGRI-0015	BSc in Agriculture (WD_SAGUL_D)	3 / 5 / M

Indicative Content

- Assessment of the value (financial and environmental) of white clover in grass swards
- Sustainable agriculture, including an overview of environmentally-sustainable farm systems, alternative methods of pest control, and the design of constructed wetland systems to control farm pollution
- An Environmental Impact Assessment of a site and the writing of an Environmental Impact Statement
- On-site and laboratory analyses of farm pollutants, including nitrates and phosphates
- Evaluation of EU and Irish environmental legislation and schemes, discussions to include reasons for implementation as well as effectiveness of legislation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Debate the value of introducing white clover into a grass sward.
2. Explore the general principles of sustainable agriculture and how they can be used to reduce the impact of farming on the environment.
3. Prepare a basic Environmental Impact Assessment of a site and construct an Environmental Impact Statement.
4. Analyse water for pollutants including nitrates, ammonia and phosphates and use the results to demonstrate the effectiveness of constructed wetland systems.
5. Examine in detail Irish and EU legislation and strategies for environmental protection at farm level, and consider how these measures could be improved.

Learning and Teaching Methods

- Lectures.
- Field trip to a constructed wetland system.
- On-site visit to facilitate the preparation of an Environmental Impact Statement.
- Laboratory practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,5
Continuous Assessment	50%	
Lab Report	50%	3,4

Assessment Criteria

<40%: Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.

40%-49%: Will be able to show a basic knowledge of the key learning outcomes including the impact of farming on the environment and be able to perform basic environmental analysis.

50%-59%: As well as the above the student will be able to demonstrate an understanding of some of the complexities of the topics and predict the impact of various farm practices on the environment. The student will also be able to perform good quality environmental analyses.

60%-69%: In addition, the student will demonstrate more detailed knowledge of all the topics including the role of sustainable agriculture and demonstrate an ability to evaluate detailed knowledge acquired in the module to new material. The student will also be able to perform very good quality environmental analysis.

70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, including the role of sustainable agriculture and show the ability to evaluate and relate topics that overlap all of the learning outcomes. The student will also be able to perform high quality environmental analysis.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	18	
Field Trips	6	
Independent Learning	87	

Essential Material

- Addiscott, T.M. _Nitrate, Agriculture and the Environment_. Oxfordshire, UK: CABI Publishing , 2005.

Supplementary Material

- Warren, J., C. Lawson and K. Belcher . _The Agri-Environment_ . UK: Cambridge University Press, 2009.

Requested Resources

- Room Type: Chemistry Lab